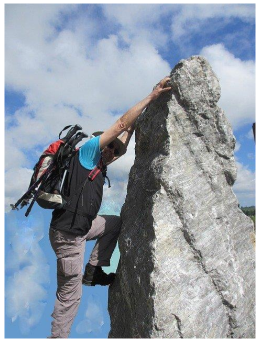
11.6.4 Wrap-Up: Reflection

1. Complete the table by reflecting on the day's lesson.

<table>
<tr><td>Today I deserve

for …</td><td></td></tr>
<tr><td>Today I

by doing …</td><td></td></tr>
<tr><td>Today I was


by …</td><td></td></tr>
</table>

Unit 11, Lesson 7: Solving Real-World Problems With Rational Numbers – Part 3

Benchmark

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

Learning Target

- ☐ I can solve problems with at least three operations involving any of the four operations with rational numbers.

11.7.1 Warm-Up: Notice and Wonder

1. What do you notice about the binder clips in the image?

2. What questions come to mind when you look at this image?

3. The stack of papers contains about 260 sheets of paper. Estimate the number of sheets each binder clip can hold.

4. To make a good estimate, what information would be useful to know?

5. What tools or resources could help you find that information?

11.7.2 Activity: Gallery Walk

Your teacher will assign your group one of the following scenarios. Work together to answer the question(s), writing neatly on the chart paper provided. Show all your work to prepare a presentation for the other students in your class.

Scenario 1	Marie plans to use a ride sharing service for the trip from Tampa International Airport to her hotel. The service has a booking fee of $\$2.45$ and a cost of $\$0.42$ for the first $\frac{1}{4}$ mile and an additional fee of $\$0.16$ for each 0.1 mile after the first $\frac{1}{4}$ mile. 1. How much will the trip cost if Marie's hotel is 7.25 miles from the airport? 2. Marie decides to stop by a grocery store before arriving at her hotel. The ride sharing service charges $\$0.45$ for each minute they have to wait on a customer. Her total bill was $\$21.72$. How long was Marie in the grocery store?

Scenario 2

Leah and her parents have an agreement. She borrowed money from them to buy a video gaming system and is supposed to pay them back within 3 months. If Leah does not have the balance paid by the agreed-upon time, she will have to give the system to her parents until it is paid off. Leah and her parents created a payment plan to help her meet this goal. Starting on July 1, Leah should pay $\$5.95$ per day until the gaming system is paid off. On July 5^{th}, her balance was $-\$10.25$ since she couldn't pay the full amount each day. Leah was not able to pay anything again for 15 days. On July 20^{th}, Leah paid $\$85.00$.

1. How much had Leah paid her parents by July 5^{th}?

2. By July 20^{th}, how much more or less had Leah paid than what was required by the payment plan?

Scenario 3

A coffee shop keeps track of changes in their coffee inventory. A week of their record-keeping is shown.

Day	Change in Inventory (lb)
Monday	150
Tuesday	-43.3
Wednesday	-50.5
Thursday	72.8
Friday	-52.7
Saturday	-21.6
Sunday	0

1. How much coffee is left at the end of the week?

2. Coffee is delivered in 50-pound (lb) bags, 3 bags at a time. How many pounds of coffee were sold on Thursday?

Scenario 4	
	Recall from the previous lesson that The Better Utility Company charges $\$0.12$ per kilowatt-hour for the energy a customer uses that is provided by the utility company. The company gives an adjustment of $-\$0.025$ for every kilowatt-hour of electricity a customer with solar panel(s) generates that they do not use themselves.

The table shows the number of kilowatt-hours of the electricity a customer used and generated each week for a month.

	Kilowatt-Hours Used	Kilowatt-Hours Generated that were not Used by the Customer
Week 1	112.08	253.2
Week 2	181.5	357.6
Week 3	101	308
Week 4	200.42	214.4

1. What is the amount due for this month?

2. The average customer's bill is $\frac{2}{3}$ of what they would have been charged if they had not produced electricity using solar panels. Explain if this customer's savings was below or above that average.

11.7.3 Your Turn!

1. Araya is making red ribbons to sell to support a local nonprofit organization. She purchased a roll of fabric that is $30\frac{3}{4}$ feet in length and each ribbon uses 0.635 feet of fabric.

   If she already has 16 ribbon orders, how many additional ribbons will she be able to sell with the single roll? Show your work.

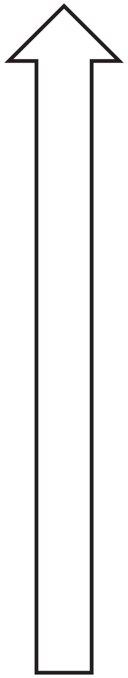
2. Joanne purchased $2\frac{1}{4}$ pounds of tomatoes, $3\frac{1}{8}$ pounds of red onions, $\frac{3}{4}$ pounds of asparagus, and 3 lemons from a grocery store.

Produce	Price
Tomatoes	$\$2.99$ per pound
Red Onions	$\$1.79$ per pound
Asparagus	$\$2.49$ per pound
Lemons	$\$0.79$ each

   Based on the listed prices, how much would Joanne spend?

11.7.4 Wrap-Up: Reflection

1. Color in the arrow up to the statement which best describes your current understanding of solving real-world problems with rational numbers.



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will make sure that I get help.

I do not understand any of this yet. There are things I could do to be a better learner.

Unit 11, Lesson 8: Evaluating Numeric Expressions – Part 1

Benchmarks

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

Learning Target

☐ I can evaluate numeric expressions with up to four operations.